UNIT 1 · TOPIC 3(D)

Planting Equipment

Syllabus topic: Equipment for land development, tillage, sowing, planting and transplanting

Row-crop planter

- Uses hill-drop or drill-plant seed plates. Plate types: **edge drop, flat drop, hill drop**.
- Plate speed affects cell fill — keep the plate peripheral speed **low** for uniform fill.
- Cell/plate metering suits row crops such as maize and cotton.

Precision (pneumatic/vacuum) planter

- Suction holds each seed on a perforated disc, giving **near-100% singulation**.
- Typical vacuum: **40–60 mbar**.
- Used for maize, sunflower, cotton and vegetables.

Inclined-plate planter

- Cells on an inclined rotating plate meter seed gently — used for groundnut because it minimises seed damage.

Crop-specific planters (potato, sugarcane, cotton, groundnut) are in Topic 7.